Predicting Tri-Level Irrigation Needs in Nepal Using a Random Forest Classification Framework

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***Abstract*—Agriculture remains a major part of Nepal's economy, yet irrigation practices in many areas are still reactive and insufficiently adapted to local soil and climatic conditions. This study presents a machine learning approach for predicting irrigation need as a three-class problem with Low, Medium, and High categories. A synthetic Nepal-calibrated agricultural dataset from kaggle containing soil, climate, crop, and farm-management variables was used. Three classification models, namely Logistic Regression, Decision Tree, and Random Forest, were compared.**

Particular emphasis was placed on data preprocessing, feature engineering, and class imbalance handling. A structured pipeline was applied, including data validation, categorical encoding, scaling of numerical variables, engineered irrigation-related features, and before-and-after SMOTE evaluation. Model performance was assessed using accuracy, macro-averaged precision, macro-averaged recall, macro-averaged F1-score, and confusion matrices. The results showed that Random Forest achieved the best overall performance, with a baseline macro F1-score of 0.9678. Although SMOTE slightly improved the Decision Tree model, it did not improve the best-performing Random Forest model and reduced the performance of Logistic Regression.

The findings show that ensemble tree-based learning is highly effective for this irrigation classification task and can provide a useful foundation for future irrigation decision-support tools in Nepal.

I. Introduction

Agriculture continues to play a central role in Nepal's economy and rural livelihoods. The sector remains closely tied to household income, food security, and regional development, while productivity is strongly influenced by water availability, seasonal rainfall, and field-level management practices [1], [2]. In many parts of Nepal, irrigation decisions are still made using traditional schedules or visible signs of crop stress. Such reactive practices can lead to delayed intervention, inefficient water use, and reduced crop productivity [3].

These challenges have become more serious under increasing climatic variability. Irregular monsoon patterns, prolonged dry periods, and unseasonal rainfall have made irrigation planning more uncertain across Nepal's farming regions. The problem is further complicated by Nepal's geographical diversity, as agricultural conditions differ substantially between the Terai plains and the hill regions. As a result, irrigation demand cannot be managed effectively through a uniform rule-based approach, creating the need for data-driven methods that reflect local soil, climate, and crop conditions [4], [3].

Machine learning provides a suitable approach because it can model complex relationships among multiple agricultural variables and convert them into predictive decision-support tools. In this study, irrigation need was formulated as a three-class classification problem with output labels of Low, Medium, and High. A Nepal-calibrated agricultural dataset was used, containing variables related to soil characteristics, climatic conditions, crop type, irrigation practice, and regional context. Three classification models were compared: Logistic Regression, Decision Tree, and Random Forest. In addition, the effect of Synthetic Minority Over-sampling Technique (SMOTE) was examined because the High irrigation need class was the minority class in the dataset.

The study makes three main contributions. First, it presents a complete and reproducible machine learning workflow for tri-level irrigation classification in a Nepal-focused agricultural setting. Second, it gives explicit importance to data preprocessing, feature engineering, and class imbalance handling rather than treating them as minor implementation details. Third, it provides an interpretable final model through confusion-matrix analysis and feature-importance analysis, thereby improving the practical usefulness of the findings.

II. Literature Review

Machine learning has increasingly been applied to agricultural decision-support tasks such as crop recommendation, irrigation automation, yield prediction, and soil health analysis. Existing studies indicate that these methods are particularly valuable when agricultural outcomes depend on interacting soil, weather, and management variables rather than on simple linear relationships [10], [9]. This is especially relevant to irrigation problems, where the required water level is influenced simultaneously by environmental conditions, crop development, and prior management actions.

Recent research further suggests that intelligent irrigation and crop-support systems can improve resource efficiency when environmental variables are combined with data-driven modelling. Earlier work has shown that adaptive irrigation systems, soil-informed recommendation models, explainable machine learning approaches, and engineered soil-related features can all strengthen agricultural prediction performance and improve trust in recommendation systems [5], [6], [7], [8].

At the methodological level, the present study is also informed by prior research on imbalanced classification and tabular supervised learning. SMOTE remains one of the most widely used approaches for minority-class oversampling [11], while broader work on learning from imbalanced data warns

that resampling does not uniformly improve all classifiers and should be evaluated empirically rather than assumed to be beneficial [12]. For structured tabular data, Random Forest continues to be one of the strongest and most interpretable ensemble methods [13], whereas Logistic Regression remains a useful linear baseline [14]. These studies together support the design choice of this paper: to compare multiple classifiers under identical preprocessing conditions and to evaluate the effect of SMOTE directly.

Despite these developments, there remains limited publicly documented work focused specifically on multi-class irrigation need prediction using a Nepal-calibrated agricultural dataset and an explicit baseline-versus-SMOTE comparison. This study addresses that gap by combining preprocessing, feature engineering, imbalance handling, comparative evaluation, and final-model interpretation within one reproducible workflow.

III. Problem Statement and Objectives

Nepal's agricultural sector remains highly dependent on seasonal rainfall and locally variable irrigation practices. In many farming areas, irrigation decisions are still made reactively, often after visible crop stress has already appeared. Such practices can result in both over-irrigation and under-irrigation, depending on local soil conditions, crop stage, and weather patterns. The absence of a simple data-driven prediction mechanism makes it difficult to estimate irrigation demand in advance, particularly across Nepal's diverse agricultural regions.

This study addresses that problem by formulating irrigation need prediction as a three-class classification task, where irrigation demand is categorized as Low, Medium, or High using soil, climate, crop, and management-related variables. The main objective of the study was to develop and evaluate machine learning models for this task and to identify the most effective model for predicting irrigation need. In addition, the effect of class imbalance handling through SMOTE was examined in order to determine whether resampling improved classification performance, particularly for the minority High irrigation need class.

IV. Dataset Description

The dataset used in this study was a synthetic agricultural dataset from kaggle which was calibrated to reflect Nepal's farming context. It contained a total of 900,000 records, divided into 630,000 training instances and 270,000 test instances. The training set contained 20 input features and one target variable, *Irrigation_Need*, while the test set contained only the input variables. The target variable was defined as a three-class label with the categories Low, Medium, and High.

The dataset included variables related to soil properties, climatic conditions, crop characteristics, irrigation practice, and regional context. These variables included soil type, soil pH, soil moisture, organic carbon, electrical conductivity, temperature, humidity, rainfall, sunlight hours, wind speed, crop type, crop growth stage, season, irrigation type, water source, field area, mulching use, previous irrigation amount, and region. The dataset was designed to reflect Nepal-specific agricultural patterns, including traditional seasons such as Barkhe, Hiunde, and Chaite, as well as the five historical development regions.

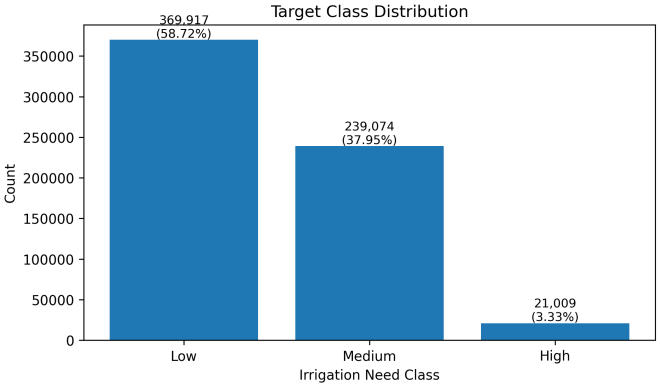

Fig. 1. Distribution of the target classes in the training dataset

Initial inspection showed that the dataset was clean and well-structured. No missing values or duplicate rows were found in either the training or test split. However, the target variable was imbalanced. In the training set, the Low class accounted for 58.72% of records, the Medium class accounted for 37.95% of records, and the High class accounted for only 3.33%. This imbalance was important because it increased the difficulty of correctly identifying the minority High irrigation need class during model training and evaluation. Figure 1 illustrates the class imbalance in the training data, where the Low and Medium classes are substantially more frequent than the High irrigation need class.

V. Methodology

A. Overview

A supervised machine learning workflow was followed to predict irrigation need as a three-class classification problem. The overall process consisted of data validation, preprocessing, feature engineering, model training, class imbalance handling, evaluation, and final model selection. Three classification models were used for comparison: Logistic Regression, Decision Tree, and Random Forest. These models were selected to represent baseline linear learning, interpretable tree-based learning, and ensemble tree-based learning.

B. Workflow Summary

The overall machine learning workflow used in this study can be summarized as follows:

$$\text{Raw Data} \rightarrow \text{Data Validation} \rightarrow \text{Preprocessing} \rightarrow \text{Feature Engineering} \rightarrow \text{Baseline Models} \rightarrow \text{SMOTE Models} \rightarrow \text{Evaluation} \rightarrow \text{Final Model Selection}$$

More specifically, the workflow consisted of raw data checking, preprocessing, engineered feature generation, baseline and SMOTE-based model development, comparative evaluation, and final model selection for test prediction.

C. Machine Learning Models

Three classification models were selected for comparative evaluation: Logistic Regression, Decision Tree, and Random Forest. These models were chosen because they represent different learning behaviours and provide a meaningful basis for comparison on structured agricultural data.

Logistic Regression was used as a baseline linear classifier. It is simple, computationally efficient, and interpretable, making it useful for establishing a reference level of performance. However, because it models linear decision boundaries, its ability to capture complex interactions among environmental and management-related variables is limited.

Decision Tree was selected because it can model non-linear relationships and produce rule-based predictions that are relatively easy to interpret. It is well suited for tabular classification tasks, but it may become unstable or overfit if not properly constrained. In this study, tree depth and split-related parameters were controlled to reduce this risk.

Random Forest was used as an ensemble tree-based classifier. It combines the predictions of multiple decision trees and is generally more robust than a single tree model. In addition to strong predictive performance on tabular datasets, Random Forest also provides feature importance estimates, which support interpretation of the final model and help identify the variables most strongly associated with irrigation need.

D. Data Preprocessing

The training and test files were first checked for structural consistency, duplicate rows, and missing values. Since no missing values or duplicate records were found, no imputation or record removal was required. The identifier column was removed before modeling because it had no predictive value. The dataset was then divided into training and validation subsets using a stratified 80:20 split so that the target-class distribution was preserved.

Categorical variables were transformed through encoding, while numerical variables were standardized within the preprocessing pipeline. This step was particularly important for Logistic Regression and the SMOTE-based setting, where distance- and scale-sensitive operations can be affected by inconsistent feature magnitudes. For the SMOTE experiment, preprocessing was adapted so that mixed numerical and categorical variables could be handled appropriately within the resampling workflow. Particular attention was given to keeping preprocessing consistent across models so that differences in performance could be attributed more confidently to the learning algorithms themselves rather than to differences in data handling.

E. Feature Engineering

Feature engineering was introduced to capture interactions among the original variables and to reflect irrigation-related relationships more explicitly. Four additional features were created:

$$\text{Moisture Deficit} = \text{Humidity} - \text{Soil Moisture} \tag{1}$$

$$\text{Soil Stress Score} = |\text{Soil pH} - 7| + \text{Electrical Conductivity} \tag{2}$$

$$\text{Temperature–Rainfall Ratio} = \frac{\text{Temperature_C}}{\text{Rainfall_mm} + 1} \tag{3}$$

$$\text{Irrigation per Hectare} = \frac{\text{Previous Irrigation_mm}}{\text{Field Area_hectare} + 1} \tag{4}$$

These features were intended to represent water deficit, soil-related irrigation stress, the relationship between atmospheric demand and rainfall input, and irrigation intensity relative to field size. The inclusion of engineered features helped strengthen the practical relevance of the model by embedding domain-informed relationships into the feature space.

F. Model Training and Imbalance Handling

Three classifiers were trained and evaluated. Logistic Regression was used as a linear baseline with a maximum iteration limit of 1000 and class weighting [14]. Decision Tree was configured with controlled depth and minimum split constraints in order to reduce overfitting, while Random Forest was trained as an ensemble model using 200 trees [13]. In the baseline setting, class-weight adjustment was used where appropriate to reduce bias toward the majority class. In the second setting, SMOTE-based resampling was applied before model training in order to examine whether minority-class representation could be improved [11], [12].

G. Evaluation Metrics

Model performance was assessed using accuracy, macro-averaged precision, macro-averaged recall, macro-averaged F1-score, and confusion matrices. Macro F1-score was treated as the main comparison metric because it gives equal importance to all three classes, including the minority High irrigation need class. The final model was selected on the basis of overall predictive performance and class balance rather than accuracy alone.

VI. Implementation and Reproducibility

The implementation was carried out in Python using a modular machine learning workflow. Jupyter Notebook was used for interactive experimentation and result inspection, while Python scripts were used for reproducible execution of the baseline models, SMOTE-based models, result comparison, feature-importance extraction, and final test prediction. Data manipulation and analysis were performed using Pandas and NumPy, model development was carried out using scikit-learn and imbalanced-learn, and visual outputs were produced using Matplotlib and Seaborn [15].

The project pipeline generated a set of experiment artifacts including model comparison summaries, classification reports, confusion matrices, the saved final Random Forest model, and

final test predictions. This implementation structure improves reproducibility and aligns the report with the actual code workflow used in the project.

VII. Results and Discussion

The comparative results showed that Random Forest achieved the strongest overall performance among all evaluated models. In the baseline setting, Random Forest produced an accuracy of 0.9850 and a macro F1-score of 0.9678, which was the highest among the three models. Decision Tree also performed well, with a macro F1-score of 0.9523, while Logistic Regression produced a considerably lower macro F1-score of 0.7435. Table I summarizes the performance of all models under both the baseline and SMOTE-based settings. These results indicate that tree-based models were more suitable for this classification task than the linear baseline model.

All precision, recall, and F1 values reported in Table I are macro-averaged metrics. The effect of SMOTE was then examined to determine whether minority-class handling improved model performance. The results showed that SMOTE did not improve all models equally. A small improvement was observed for Decision Tree, where the macro F1-score increased from 0.9523 to 0.9553. However, the performance of Random Forest decreased slightly after SMOTE, with macro F1-score falling from 0.9678 to 0.9637. Logistic Regression also performed worse after SMOTE, with its macro F1-score decreasing from 0.7435 to 0.6224. These findings suggest that although resampling increased representation of the minority class, it did not necessarily lead to better overall generalization.

A closer inspection of the class-wise results showed that Logistic Regression achieved relatively high recall for the High irrigation need class, but this came at the cost of weaker balance across the full classification task. In contrast, Random Forest maintained strong and stable performance across all three classes, which explains its superior macro F1-score. Since macro F1-score gives equal importance to each class, including the minority class, it was treated as the main selection criterion. On this basis, the baseline Random Forest model was selected as the final model.

Overall, the results indicate that irrigation need prediction in this dataset was better captured by nonlinear tree-based learning than by linear classification. The strong performance of Random Forest also suggests that the interaction between soil, crop, climate, and irrigation-related variables was complex and not easily represented by a simple linear boundary. While SMOTE was useful as a comparative imbalance-handling method, the final findings showed that the best-performing model remained the baseline Random Forest trained on the original dataset.

TABLE I
Validation performance of baseline and SMOTE-based models

Setting	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Baseline	Logistic Regression	0.8430	0.7059	0.8509	0.7435
Baseline	Decision Tree	0.9815	0.9394	0.9671	0.9523
Baseline	Random Forest	0.9850	0.9680	0.9677	0.9678
SMOTE	Logistic Regression	0.7087	0.5893	0.7401	0.6224
SMOTE	Decision Tree	0.9824	0.9476	0.9638	0.9553
SMOTE	Random Forest	0.9833	0.9619	0.9656	0.9637

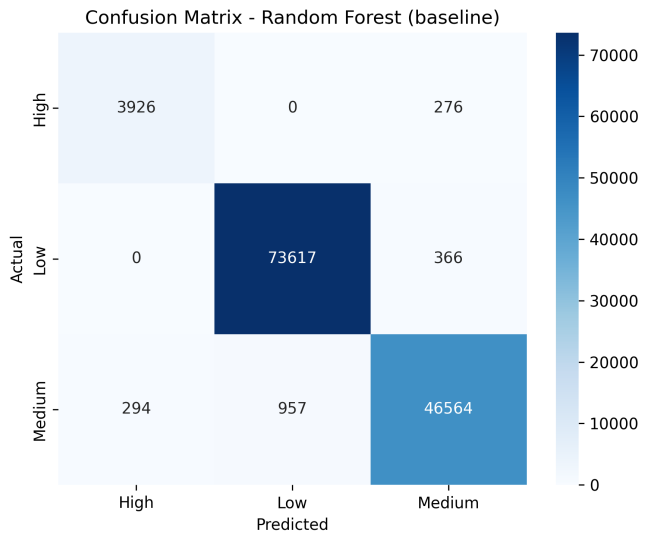

Fig. 2. Confusion matrix for the baseline Random Forest model on the validation set

A. Interpretation of the Final Model

Figure 2 shows the class-wise prediction performance of the final baseline Random Forest model. Strong classification performance was observed across all three irrigation levels, with particularly high consistency for the Low and Medium classes. The model also performed well on the minority High irrigation need class, which supports its selection as the final model.

To better understand the behaviour of the final selected model, feature-importance analysis was carried out using the fitted baseline Random Forest classifier. Table II presents the five most influential predictors identified by the model. The results showed that Soil_Moisture was the most important feature, followed by Temperature_C, Wind_Speed_kmh, Rainfall_mm, and Crop_Growth_Stage_Harvest. These findings are consistent with the agricultural logic of irrigation demand, since water requirement is strongly influenced by existing soil water availability, atmospheric conditions, and crop development stage.

The strong performance of Random Forest suggests that irrigation need in this dataset was determined by nonlinear interactions among environmental and management-related variables rather than by simple linear relationships. In particular, the combined effect of low soil moisture, reduced rainfall, higher temperature, and stronger drying conditions likely contributed to stronger separation of the High irrigation need class. This also helps explain why Random Forest

TABLE II
TOP FIVE FEATURE IMPORTANCES FROM THE FINAL RANDOM FOREST MODEL

Feature	Importance
Soil_Moisture	0.2406
Temperature_C	0.1254
Wind_Speed_kmh	0.0963
Rainfall_mm	0.0706
Crop_Growth_Stage_Harvest	0.0690

outperformed Logistic Regression, as the latter is less capable of capturing complex feature interactions.

B. Practical Implications

From an application perspective, these findings indicate that irrigation prediction in Nepal should not rely on a single indicator alone. Instead, multiple factors should be considered together, especially soil condition, recent irrigation history, climatic conditions, and crop stage. A model of this kind could support more informed irrigation scheduling, guide extension workers in advising farmers, and provide a basis for future decision-support tools in precision agriculture. The final model therefore offers not only strong predictive performance, but also practical value for irrigation planning in water-stressed agricultural settings.

VIII. ETHICAL, LEGAL, AND PROFESSIONAL CONSIDERATIONS

Although the final model achieved strong predictive performance, its use should be approached responsibly. The dataset used in this study was downloaded from kaggle, which means the model may not fully represent the diversity and uncertainty of real farming conditions in Nepal. If such a model were used without validation on real field data, there would be a risk of producing misleading irrigation recommendations. In practice, incorrect predictions could contribute either to under-irrigation, which may reduce crop yield, or over-irrigation, which may waste water and affect soil quality.

From a professional perspective, the model should therefore be treated as a decision-support tool rather than a replacement for farmer judgment or agricultural expertise. Local conditions, crop knowledge, and field observation should still be considered before applying any recommendation. In addition, any real deployment would need to consider data quality, transparency of recommendations, and fair access for smallholder farmers, especially in regions with limited technical infrastructure.

IX. LIMITATIONS

Several limitations should be acknowledged in this study. First, the dataset used for model development was synthetic rather than collected directly from field observations. Although it was designed to reflect Nepal's agricultural context, it cannot fully represent real-world variability, measurement noise, and local anomalies. As a result, the reported model performance should be interpreted with caution, since results on actual field data may differ.

Second, the dataset did not fully reflect the true geographic distribution of crops across Nepal. The crop–region patterns were more uniform than would normally be expected in practice, which may have introduced distributional bias into the modeling process. In addition, the average field size represented in the dataset was larger than the typical size of smallholder farms in Nepal. This may limit the generalisability of the findings, especially in cases where irrigation behaviour is influenced by farm size and local resource constraints.

Finally, the dataset had no temporal structure, meaning that irrigation changes across time and across growing periods could not be modelled directly. The study was therefore limited to static tabular classification rather than sequential irrigation forecasting. The modeling process also did not include economic considerations such as the cost of pumping water or fuel use, even though such factors may influence real irrigation decisions in practice.

X. CONCLUSION

This study presented a machine learning approach for predicting irrigation need in Nepal as a three-class classification problem. Three models, namely Logistic Regression, Decision Tree, and Random Forest, were developed and evaluated using a structured preprocessing pipeline and two experimental settings: baseline training and SMOTE-based imbalance handling. The results showed that Random Forest achieved the strongest overall performance, with the highest macro F1-score among all evaluated models.

The comparative analysis also showed that SMOTE did not improve all models equally. A slight improvement was observed for Decision Tree, whereas the performance of Random Forest and Logistic Regression decreased after resampling. Based on the overall results, the baseline Random Forest model was selected as the final model. The study contributes a reproducible machine learning workflow for Nepal-specific irrigation need prediction, a direct baseline-versus-SMOTE comparison, and an interpretable final Random Forest model supported by confusion-matrix and feature-importance analysis. These findings indicate that tree-based ensemble learning was the most effective approach for this irrigation classification task and provided the best balance across all three classes.

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